QUESTION 3

PROBLEM STATEMENT

Create a program that allocates dynamic memory to store 15 integers .It should calculate and display the sum and average of this numbers and free the allocated space .

ALGORITHIM

Step 1: Start.

Step 2: Declare a pointer for integers.

Step 3: Allocate memory for 15 integers using malloc().

Step 4: Enter the 15 integers.

Step 5: Calculate the total of the integers.

Step 6: Calculate the average.

Step 7: Display the total and average.

Step 8: Free the allocated memory.

Step 9: Stop.

SOURCE CODE

```
#include <stdio.h>

#include <stdlib.h>

int main() {

  int *arr;

  int sum = 0;

  float avg;

  arr = (int*)malloc(15 * sizeof(int));

  if (arr == NULL) {

    printf("Memory allocation failed!\n");

    return 1;

  }
```

```c
    printf("Enter 15 numbers:\n");
    for (int i = 0; i < 15; i++) {
        printf("Number %d: ", i + 1);
        scanf("%d", &arr[i]);
    }
    for (int i = 0; i < 15; i++) {
        sum = sum + arr[i];
    }

    avg = sum / 15.0;
    printf("\nSum = %d\n", sum);
    printf("Average = %.2f\n", avg);
    free(arr);
    return 0;
}
```